

Mechanistic Interpretability - 2

CS60216: Safety Fundamentals for Generative AI

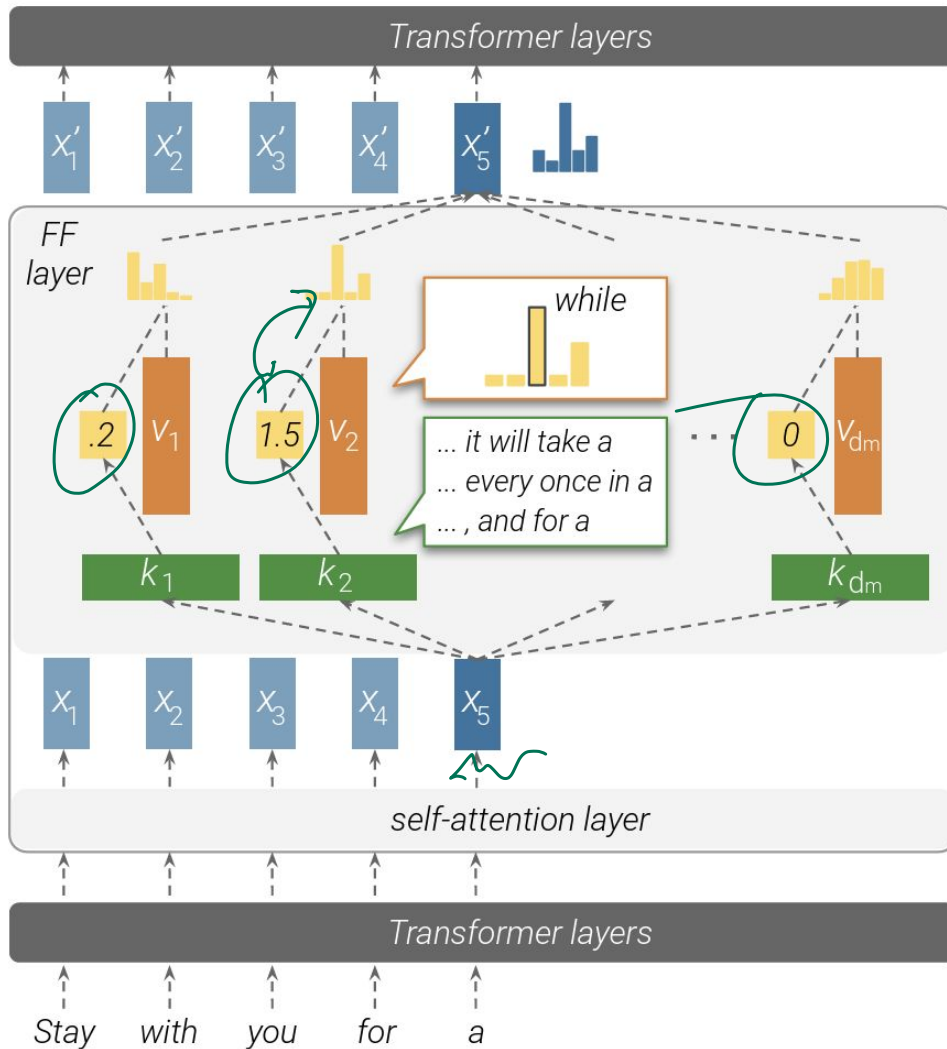
Transformer Feed-Forward Layers Are Key-Value Memories

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An illustration of how a feed-forward layer emulates a key-value memory.

Input vectors (here, x_5) are multiplied by keys to produce memory coefficients (e.g., the memory coefficient for v_1 is 0.2), which then weigh distributions over the output vocabulary, stored in the values.

The feed-forward layer's output is thus the weighted sum of its values.

MLPs as Key-value memories

Let $\mathbf{x} (\mathbb{R}^d)$ be a vector corresponding to some input text prefix. We can express the feed-forward layer as follows

$$\text{FF}(\mathbf{x}) = f(\mathbf{x} \cdot K^\top) \cdot V$$

Handwritten annotations:
- A green arrow points from $K^\top$ to w_1 .
- A green arrow points from V to w_2 .
- A green arrow points from the output of f to ga-out .

Here, $K, V \in \mathbb{R}^{\underline{d_m \times d}}$ are parameter matrices, and $\underline{f}$ is a non-linearity such as ReLU.

Neural Memory

Neural memory A neural memory (Sukhbaatar et al., 2015) consists of d_m key-value pairs, which we call memories.² Each key is represented by a d -dimensional vector $\mathbf{k}_i \in \mathbb{R}^d$, and together form the parameter matrix $K \in \mathbb{R}^{d_m \times d}$; likewise, we define the value parameters as $V \in \mathbb{R}^{d_m \times d}$. Given an input vector $\mathbf{x} \in \mathbb{R}^d$, we compute a distribution

Neural Memory

over the keys, and use it to compute the expected value:

$$\underbrace{p(k_i \mid x)} \propto \underbrace{\exp(\mathbf{x} \cdot \mathbf{k}_i)}$$
$$\text{MN}(\mathbf{x}) = \sum_{i=1}^{d_m} \underbrace{p(k_i \mid x)} \underbrace{\mathbf{v}_i}$$

With matrix notation, we arrive at a more compact formulation:

$$\text{MN}(\mathbf{x}) = \underbrace{\text{softmax}(\mathbf{x} \cdot K^\top)} \cdot V$$

Almost identical!

$$\underbrace{\text{FF}(\mathbf{x}) = f(\mathbf{x} \cdot K^\top) \cdot V}$$

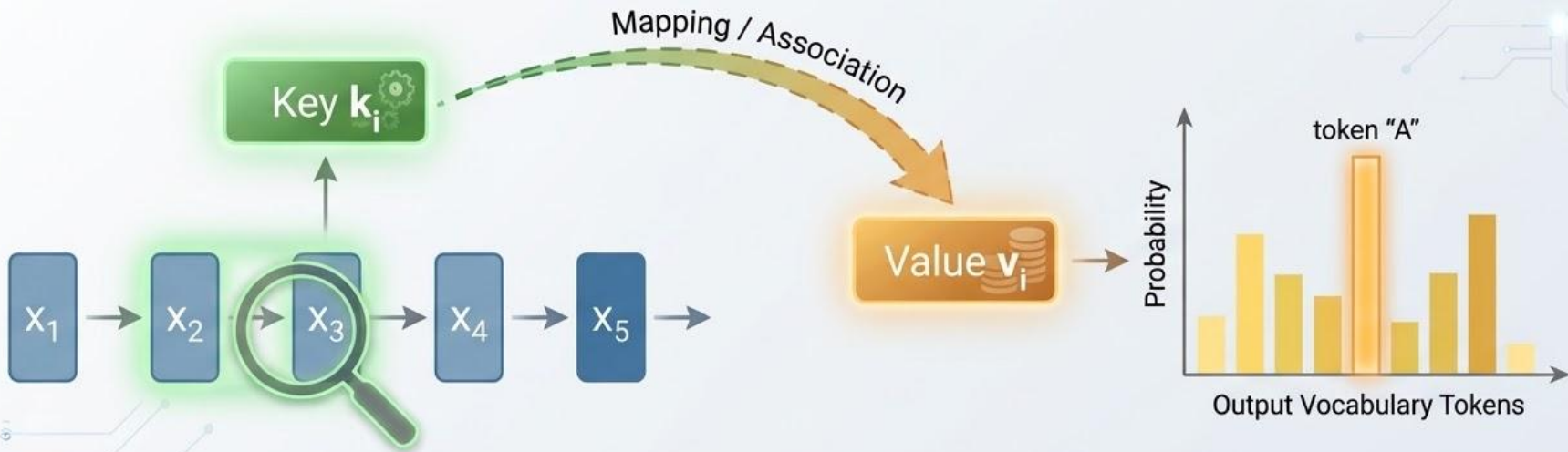
Feed-forward layers emulate neural memory

The *hidden dimension* d_m is essentially the number of memories in the layer, and the activation $\mathbf{m} = f(\mathbf{x} \cdot K^\top)$, commonly referred to as the *hidden layer*, is a vector containing an unnormalized non-negative coefficient for each memory. We refer to each $\mathbf{m}_i$ as the memory coefficient of the i th memory cell.

Conjecture from the paper

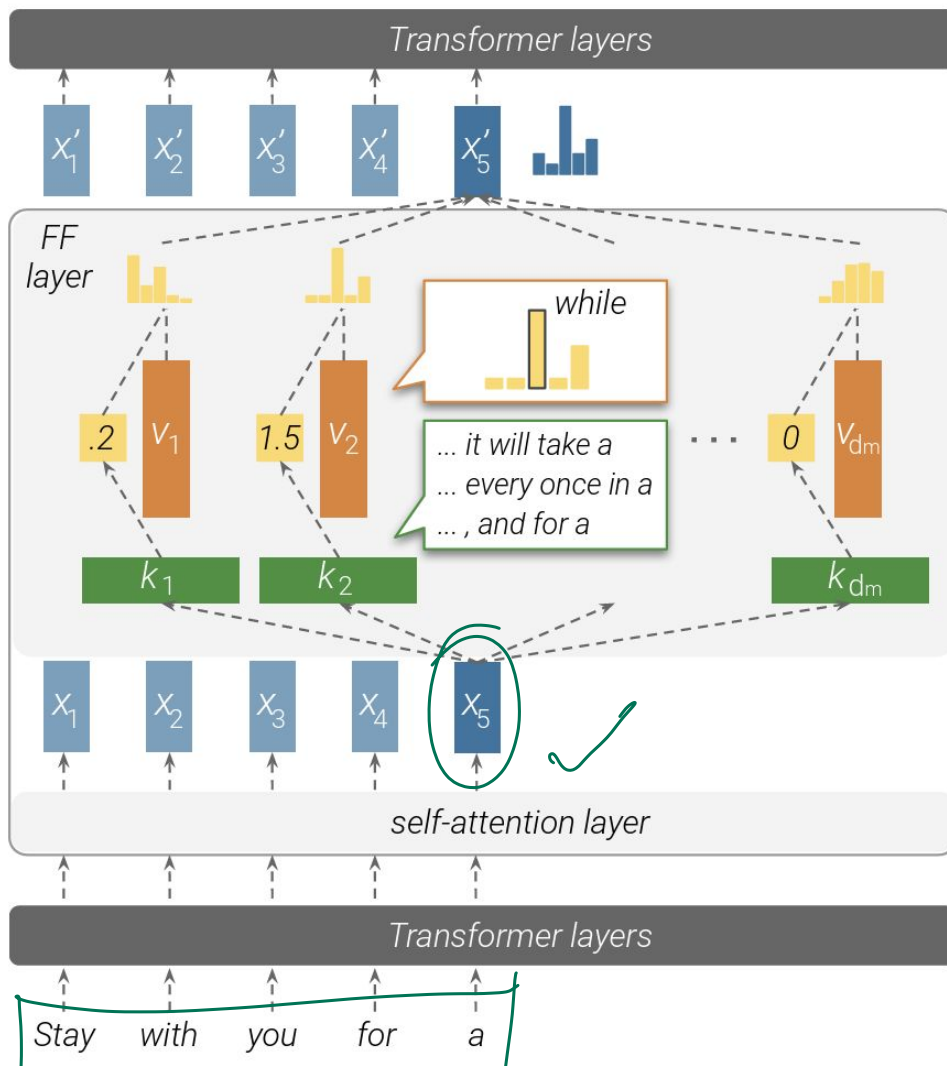
Key Vectors (k_i): Pattern Capture

Value Vectors (v_i): Next-Token Distribution



Each key vector k_i is tuned to detect a specific pattern (or set of patterns) within the input sequence.

The corresponding value vector v_i stores the probability distribution of tokens that typically follow the detected pattern.



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Experiments to support the conjecture

We posit that the key vectors K in feed-forward layers act as pattern detectors over the input sequence, where each individual key vector k_i corresponds to a specific pattern over the input prefix $x_1, \dots, x_j$.

We analyze the keys of a trained language model's feed-forward layers.

We retrieve the training examples (prefixes of a sentence) most associated with a given key, that is, the input texts where the memory coefficient is highest.

How to retrieve trigger examples?

Given

a key $\mathbf{k}_i^\ell$ that corresponds to the i -th hidden dimension of the ℓ -th feed-forward layer, we compute the memory coefficient $\text{ReLU}(\mathbf{x}_j^\ell \cdot \mathbf{k}_i^\ell)$ for every prefix $\langle x_1, \dots, x_j \rangle$ of every sentence from the WikiText-

For example, for the hypothetical sentence “I love dogs”, we will compute three coefficients, for the prefixes “I”, “I love”, and “I love dogs”.

Then, we retrieve the top- t trigger examples, that is, the t prefixes whose representation at layer yielded the highest inner product with $\mathbf{k}_i^\ell$.

Pattern Analysis

We let human experts (NLP graduate students) annotate the top-25 prefixes retrieved for each key, and asked them to

- (a) identify repetitive patterns that occur in at least 3 prefixes (which would strongly indicate a connection to the key, as this would unlikely happen if sentences were drawn at random)
- b) describe each recognized pattern, and
- (c) classify each recognized pattern as “shallow” (e.g. recurring n-grams) or “semantic” (recurring topic).

Examples of human-identified patterns

Key	Pattern	Example trigger prefixes
k_{449}^1	Ends with “ <i>substitutes</i> ” (<u>shallow</u>)	<i>At the meeting, Elton said that “for artistic reasons there could be no substitutes</i> <i>In German service, they were used as substitutes</i> <i>Two weeks later, he came off the substitutes</i>
k_{2546}^6	<u>Military</u> , ends with “ <i>base</i> ”/“ <i>bases</i> ” (<u>shallow</u> + <u>semantic</u>)	<i>On 1 April the SRSG authorised the SADF to leave their bases</i> <i>Aircraft from all four carriers attacked the Australian base</i> <i>Bombers flying missions to Rabaul and other Japanese bases</i>
k_{2997}^{10}	a “part of” relation (<u>semantic</u>)	<i>In June 2012 she was named as one of the team that competed</i> <i>He was also a part of the Indian delegation</i> <i>Toy Story is also among the top ten in the BFI list of the 50 films you should</i>
k_{2989}^{13}	Ends with a time range (<u>semantic</u>)	<i>Worldwide, most tornadoes occur in the late afternoon, between 3 pm and 7</i> <i>Weekend tolls are in effect from 7:00 pm Friday until</i> <i>The building is open to the public seven days a week, from 11:00 am to</i>
k_{1935}^{16}	TV shows (<u>semantic</u>)	<i>Time shifting viewing added 57 percent to the episode’s</i> <i>The first season set that the episode was included in was as part of the</i> <i>From the original NBC daytime version , archived</i>

Are these consistent across layers?

The lower layers (layers 1-9) are dominated by shallow patterns, often with prefixes that share the last word.

In contrast, the upper layers (layers 10-16) are characterized by more semantic patterns, with prefixes from similar contexts but without clear surface-form similarities.

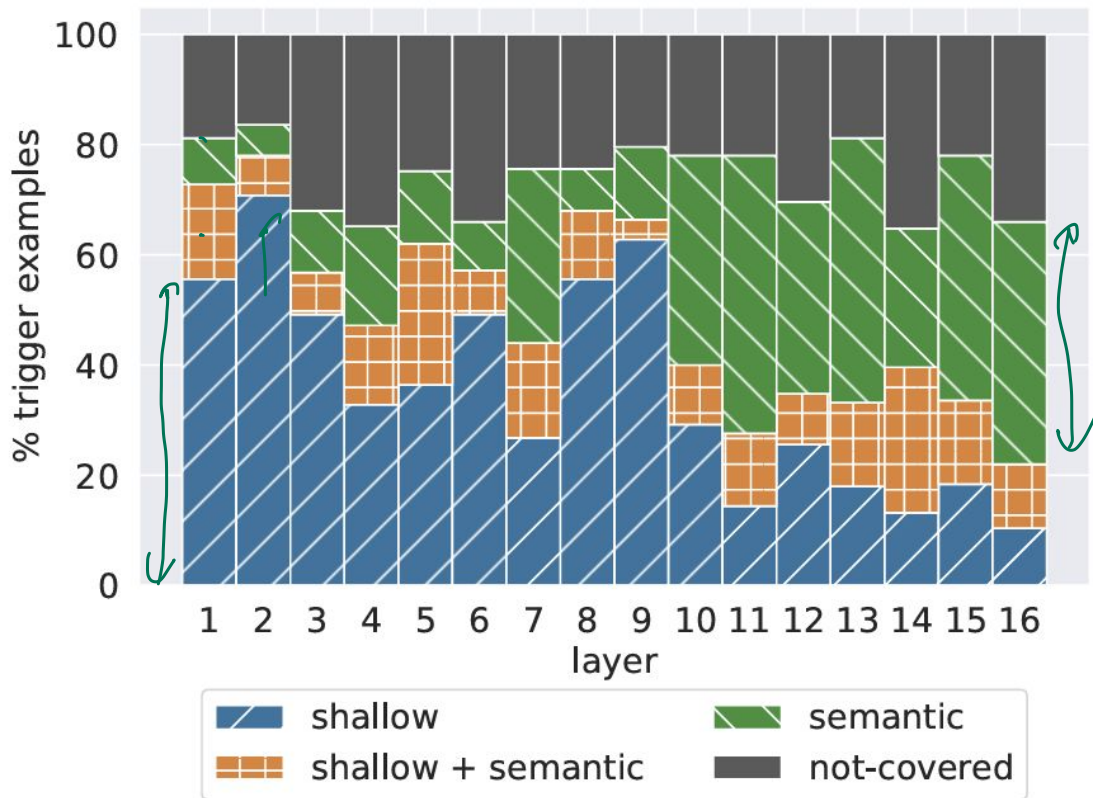


Figure 2: Breakdown of the labels experts assigned to trigger examples in each layer. Some examples were not associated with any pattern (“not-covered”).

What about values?

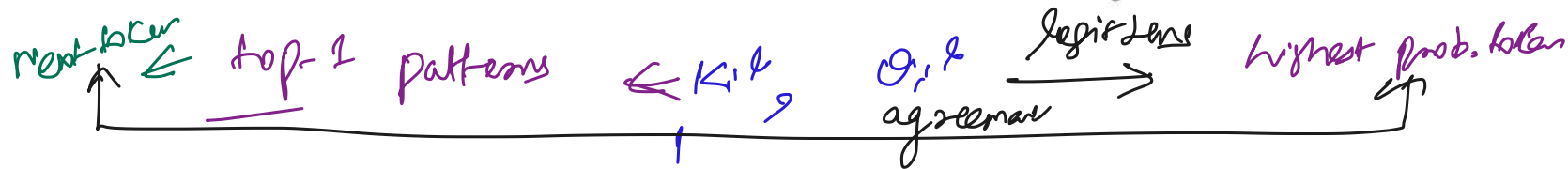
Casting values as distributions over the vocabulary. We begin by converting each value vector $\mathbf{v}_i^\ell$ into a probability distribution over the vocabulary by multiplying it by the output embedding matrix E and applying a softmax:⁴

$$\mathbf{p}_i^\ell = \text{softmax}(\underbrace{\mathbf{v}_i^\ell}_{\downarrow} \cdot \underbrace{E}_{\downarrow}).$$

K_i^ℓ : patterns
 $\mathbf{v}_i^\ell$: token $\mathbf{p}_i^\ell$ representation

Values represent distributions

Value predictions follow key patterns in upper layers. For every layer ℓ and memory dimension i , we compare the top-ranked token according to $\mathbf{v}_i^\ell$, ($\text{argmax}(\mathbf{p}_i^\ell)$) to the next token w_i^ℓ in the top-1 trigger example according to $\mathbf{k}_i^\ell$ (the example whose memory coefficient for $\mathbf{k}_i^\ell$ is the highest).



Agreement rate

Agreement rate is close to zero in the lower layers (1-10), but starting from layer 11, the agreement rate quickly rises until 3.5%.

This value is orders of magnitude higher than a random token prediction from the vocabulary, which would produce a far lower agreement rate (0.0004%)

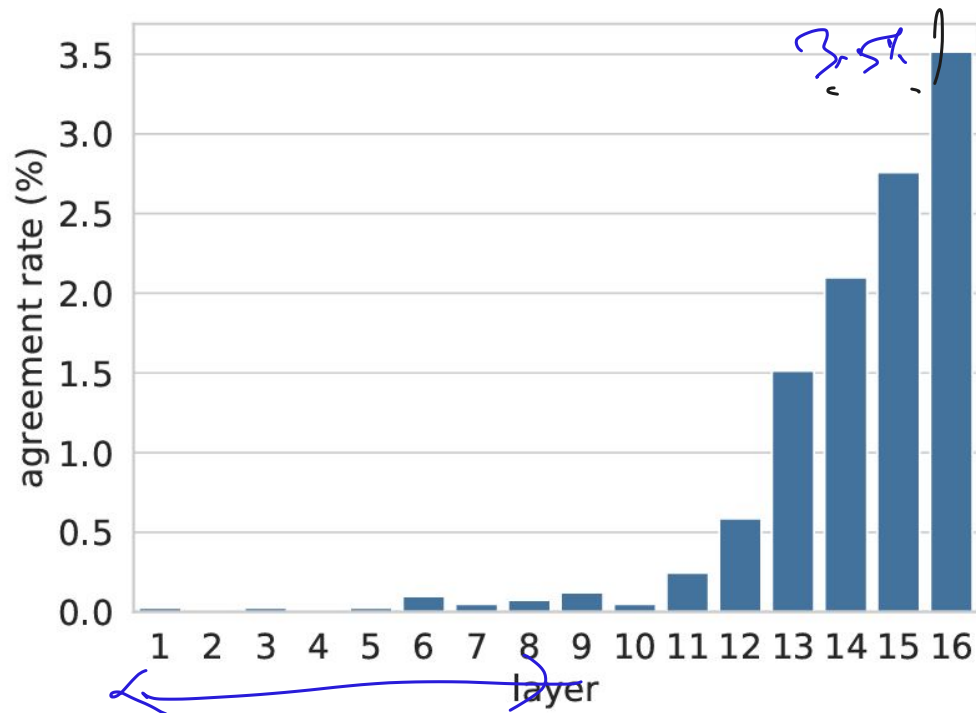


Figure 4: Agreement rate between the top-ranked token based on the value vector $\mathbf{v}_i^\ell$, and the next token of the top-ranked trigger example associated with the key vector $\mathbf{k}_i^\ell$.

Value	Prediction	Precision@50	Trigger example
v_{222}^{15}	<u>each</u>	68%	But when bees and wasps resemble <u>each</u> ^{Palham} _{next word}
v_{752}^{16}	played	16%	Her first role was in Vijay Lalwani's psychological thriller <i>Karthik Calling Karthik</i> , where Padukone was cast as the supportive girlfriend of a depressed man (<u>played</u>)
v_{2601}^{13}	extratropical	4%	Most of the winter precipitation is the result of synoptic scale, low pressure weather systems (large scale storms such as <u>extratropical</u>)
v_{881}^{15}	part	92%	Comet served only briefly with the fleet, owing in large <u>part</u>
v_{2070}^{16}	line	84%	Sailing from Lorient in October 1805 with one ship of the <u>line</u>
v_{3186}^{12}	jail	4%	On May 11, 2011, four days after scoring 6 touchdowns for the Slaughter, Grady was sentenced to twenty days in <u>jail</u>

Table 2: Example values, their top prediction, the fraction of their key's top-50 trigger examples that agree with their prediction, and a matching trigger example (with the target token marked in blue).